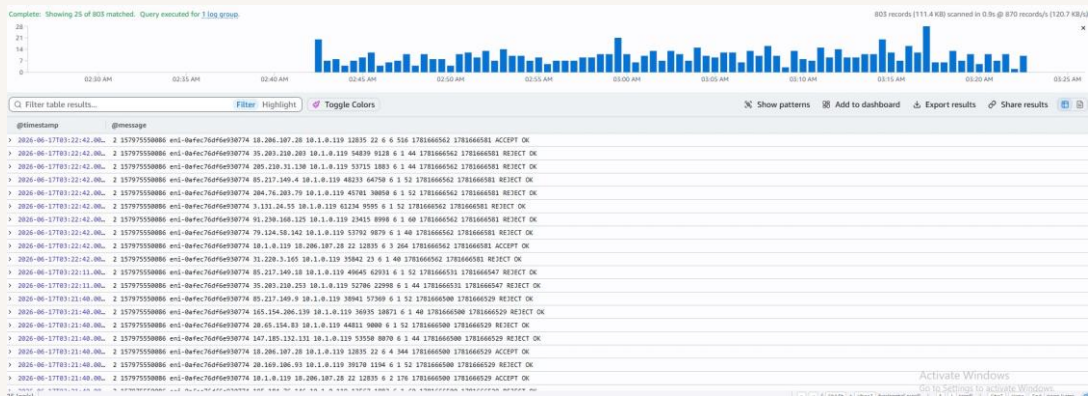


VPC Monitoring with Flow Logs

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Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a foundational AWS service that lets me control the underlying network for resources so I can control traffic flow, monitor for security, and organize my resources.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to learn how to troubleshoot VPC peering connectivity issues. I also learned how to monitor network traffic using VPC flow logs.

One thing I didn't expect in this project was...

One thing I did not expect in this project was Log Insights and all the built-in queries that give me many options on how I would like to analyze my network traffic.

This project took me...

This project took me 1.5 hours to complete.

In the first part of my project...

Step 1 - Set up VPCs

In this step, I will set up two VPCs from scratch within minutes. Network monitoring can still be done with just a single VPC, but I want to implement VPC peering as well.

Step 2 - Launch EC2 instances

In this step, I will launch two EC2 instances and place each one in its respective VPC. I am doing this so they can communicate with each other later in the project.

Step 3 - Set up Logs

In this step, I am going to set up VPC flow logs to start monitoring traffic, as well as set up a storage space.

Step 4 - Set IAM permissions for Logs

In this step, I will enable VPC Flow Logs with permission to write and send logs to CloudWatch.

Multi-VPC Architecture

I started my project by launching two VPCs. I created two public subnets, one subnet per VPC, with no private subnets.

The CIDR blocks for VPC 1 and VPC 2 are 10.1.0.0/16 and 10.2.0.0/16, respectively. They must be unique so that the IPv4 addresses of their resources don't overlap. Overlapping IP addresses can cause routing conflicts and connectivity issues.

I also launched EC2 instances in each subnet

My EC2 instances' security groups allow SSH traffic and ICMP type traffic. This is because EC2 Instance Connect will need to access our EC2 instance using SSH type traffic, and because we need to allow ICMP type traffic for connectivity tests later.



Logs

Logs record everything that happens, from users logging in to errors popping up. They are the go-to place to understand what is going on with your systems, troubleshoot problems, and keep an eye on who is doing what.

Log groups are where logs from the same source or application are stored together. Keep in mind that logs are also region-specific, meaning log data is created and saved in the region where it originated.



IAM Policy and Roles

I created an IAM policy so that I can define a rule that allows policyholders the ability to create log streams and upload them to CloudWatch, so I can analyze them later.

I also created an IAM role because services like VPC Flow Logs must be associated with a role instead of a policy. Creating an IAM role is necessary to give our VPC Flow Logs the access they need to record and upload logs.

A custom trust policy is a specific type of policy used for defining who or what is allowed to assume the IAM role.

Custom trust policy

Create a custom trust policy to enable others to perform actions in this account.

```
1 {  
2     "Version": "2012-10-17",  
3     "Statement": [  
4         {  
5             "Sid": "Statement1",  
6             "Effect": "Allow",  
7             "Principal": {"Service": "vpc-flow-logs.amazonaws.com"},  
8             "Action": "sts:AssumeRole"  
9         }  
10    ]  
11 }  
12 }
```

In the second part of my project...

Step 5 - Ping testing and troubleshooting

In this step, I will generate network traffic. This will become important when I am communicating about CloudWatch, cloud networking, or cloud engineering.

Step 6 - Set up a peering connection

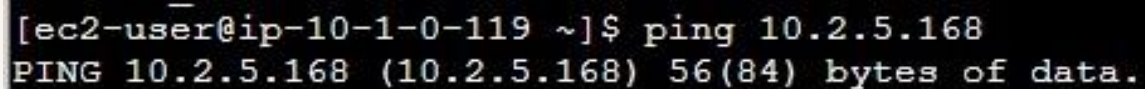
In this step, I will set up a VPC peering connection to allow both VPCs to communicate directly with each other.

Step 7 - Analyze flow logs

In this step, I will track the network data that was collected on my VPCs and then analyze the data to extract insights.

Connectivity troubleshooting

My first ping test between my EC2 instances had no replies, which means ICMP traffic could be blocked by the security group or network ACL. Maybe our traffic is being routed to the wrong path.

A terminal window with a black background and white text. The prompt is [ec2-user@ip-10-1-0-119 ~]\$. The command entered is ping 10.2.5.168. The output is PING 10.2.5.168 (10.2.5.168) 56(84) bytes of data. There is a small white cursor at the end of the output line.

```
[ec2-user@ip-10-1-0-119 ~]$ ping 10.2.5.168
PING 10.2.5.168 (10.2.5.168) 56(84) bytes of data.
```

I could receive ping replies if I ran the ping test using the other instance's public IP address, which means the network ACL and security group are properly working. Our second instance is allowing ICMP traffic.

Connectivity troubleshooting

Looking at VPC 1's route table, I identified that the ping test with Instance 2's private address failed because I do not have a route in the VPC's route table that directs traffic from one VPC to another.

To solve this, I set up a peering connection between my VPCs

I also updated both VPCs' route tables so that traffic for the other VPC can be directed to the peering connection instead of the public internet.



Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-03e2fb8549093f21	Active	No	Create Route
10.1.0.0/16	local	Active	No	Create Route Table
10.2.0.0/16	pcx-0d6cafdc8fd51860	Active	No	Create Route

Connectivity troubleshooting

I received ping replies from Instance 2's private IP address! This means setting up the peering connection and updating the route table solved the connectivity issue of our VPC traffic not being able to navigate from one VPC to another.

```

21 packets transmitted, 21 received, 0% packet loss, time 20788ms
rtt min/avg/max/mdev = 0.306/0.332/0.360/0.014 ms
[ec2-user@ip-10-1-0-119 ~]$ ping10.2.5.168
-bash: ping10.2.5.168: command not found
[ec2-user@ip-10-1-0-119 ~]$ ping 10.2.5.168
PING 10.2.5.168 (10.2.5.168) 56(84) bytes of data.
64 bytes from 10.2.5.168: icmp_seq=1 ttl=127 time=0.697 ms
64 bytes from 10.2.5.168: icmp_seq=2 ttl=127 time=0.267 ms
64 bytes from 10.2.5.168: icmp_seq=3 ttl=127 time=0.297 ms
64 bytes from 10.2.5.168: icmp_seq=4 ttl=127 time=0.279 ms
64 bytes from 10.2.5.168: icmp_seq=5 ttl=127 time=0.292 ms
64 bytes from 10.2.5.168: icmp_seq=6 ttl=127 time=0.237 ms
64 bytes from 10.2.5.168: icmp_seq=7 ttl=127 time=0.234 ms
64 bytes from 10.2.5.168: icmp_seq=8 ttl=127 time=0.237 ms
64 bytes from 10.2.5.168: icmp_seq=9 ttl=127 time=0.246 ms
64 bytes from 10.2.5.168: icmp_seq=10 ttl=127 time=0.227 ms
64 bytes from 10.2.5.168: icmp_seq=11 ttl=127 time=0.248 ms
^C
--- 10.2.5.168 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10389ms
rtt min/avg/max/mdev = 0.227/0.296/0.697/0.128 ms
[ec2-user@ip-10-1-0-119 ~]$ █

```

Analyzing flow logs

Flow logs tell us about the source and destination of the network traffic, the amount of data being transferred, and whether the traffic was accepted or rejected.

For example, the flow log I've captured tells us that traffic went from 18.206.107.28 to 10.1.0.119. We can also extract that the traffic was accepted by the security groups and the network ACLs of my VPC.

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Q Filter events - press enter to search

▶	Timestamp	Message
		There are older events to load. Load more .
▼	2026-06-17T03:05:12.000Z	2 157975550086 eni-0afec76df6e930774 18.206.107.28 10.1.0.119 12835 22 6 4 344 1781665512 1781665531 ACCEPT OK
	2 157975550086 eni-0afec76df6e930774 18.206.107.28 10.1.0.119 12835 22 6 4 344 1781665512 1781665531 ACCEPT OK	
▼	2026-06-17T03:05:12.000Z	2 157975550086 eni-0afec76df6e930774 10.1.0.119 18.206.107.28 22 12835 6 2 176 1781665512 1781665531 ACCEPT OK
	2 157975550086 eni-0afec76df6e930774 10.1.0.119 18.206.107.28 22 12835 6 2 176 1781665512 1781665531 ACCEPT OK	
▼	2026-06-17T03:05:12.000Z	2 157975550086 eni-0afec76df6e930774 31.220.3.165 10.1.0.119 35842 23 6 1 40 1781665512 1781665531 REJECT OK
	2 157975550086 eni-0afec76df6e930774 31.220.3.165 10.1.0.119 35842 23 6 1 40 1781665512 1781665531 REJECT OK	

Logs Insights

Logs Insights is a tool inside CloudWatch that helps with analyzing logs and creating visual graphs and charts through queries.

I ran the query for the top 25 recent log events. This query analyzes the flow logs collected on EC2 Instance 1 and returns the top 25 pairs of IP addresses based on the amount of data transferred between them.

Complete: Showing 25 of 803 matched. Query executed for 1 log group

803 records (111.4 KB) scanned in 0.9s @ 870 records/s (1202.7 KB/s)



Q Filter table results...

Filter Highlight

Toggle Colors

Show patterns Add to dashboard Export results Share results

@timestamp	@message
> 2026-06-17T03:22:42.000...	2 1579755500000 enl-Bafec76dffe930774 18.206.107.20 10.1.0.119 12835 22 6 6 534 1781666562 1781666581 ACCEPT OK
> 2026-06-17T03:22:42.000...	2 1579755500000 enl-Bafec76dffe930774 35.203.210.203 10.1.0.119 54839 9120 6 1 44 1781666562 1781666581 REJECT OK
> 2026-06-17T03:22:42.000...	2 1579755500000 enl-Bafec76dffe930774 205.210.31.130 10.1.0.119 53715 1883 6 1 44 1781666562 1781666581 REJECT OK
> 2026-06-17T03:22:42.000...	2 1579755500000 enl-Bafec76dffe930774 85.217.149.4 10.1.0.119 48239 64750 6 1 52 1781666562 1781666581 REJECT OK
> 2026-06-17T03:22:42.000...	2 1579755500000 enl-Bafec76dffe930774 204.76.203.79 10.1.0.119 45781 30050 6 1 52 1781666562 1781666581 REJECT OK
> 2026-06-17T03:22:42.000...	2 1579755500000 enl-Bafec76dffe930774 3.131.24.55 10.1.0.119 61234 9595 6 1 52 1781666562 1781666581 REJECT OK
> 2026-06-17T03:22:42.000...	2 1579755500000 enl-Bafec76dffe930774 91.230.160.125 10.1.0.119 23425 8998 6 1 40 1781666562 1781666581 REJECT OK
> 2026-06-17T03:22:42.000...	2 1579755500000 enl-Bafec76dffe930774 79.124.58.142 10.1.0.119 53702 9079 6 1 40 1781666562 1781666581 REJECT OK
> 2026-06-17T03:22:42.000...	2 1579755500000 enl-Bafec76dffe930774 10.1.0.119 10.200.107.20 22 12835 6 3 264 1781666562 1781666581 ACCEPT OK
> 2026-06-17T03:22:42.000...	2 1579755500000 enl-Bafec76dffe930774 31.220.3.165 10.1.0.119 35042 23 6 1 40 1781666562 1781666581 REJECT OK
> 2026-06-17T03:22:11.000...	2 1579755500000 enl-Bafec76dffe930774 85.217.149.10 10.1.0.119 49645 62951 6 1 52 1781666531 1781666547 REJECT OK
> 2026-06-17T03:22:11.000...	2 1579755500000 enl-Bafec76dffe930774 35.203.210.253 10.1.0.119 52706 22990 6 1 44 1781666531 1781666547 REJECT OK
> 2026-06-17T03:21:40.000...	2 1579755500000 enl-Bafec76dffe930774 85.217.149.9 10.1.0.119 38941 57369 6 1 52 1781666500 1781666529 REJECT OK
> 2026-06-17T03:21:40.000...	2 1579755500000 enl-Bafec76dffe930774 165.154.206.139 10.1.0.119 36935 10871 6 1 40 1781666500 1781666529 REJECT OK
> 2026-06-17T03:21:40.000...	2 1579755500000 enl-Bafec76dffe930774 20.65.154.83 10.1.0.119 44811 9000 6 1 52 1781666500 1781666529 REJECT OK
> 2026-06-17T03:21:40.000...	2 1579755500000 enl-Bafec76dffe930774 147.180.132.131 10.1.0.119 39550 8079 6 1 44 1781666500 1781666529 REJECT OK
> 2026-06-17T03:21:40.000...	2 1579755500000 enl-Bafec76dffe930774 18.206.107.20 10.1.0.119 12835 22 6 4 364 1781666500 1781666529 ACCEPT OK
> 2026-06-17T03:21:40.000...	2 1579755500000 enl-Bafec76dffe930774 20.169.106.93 10.1.0.119 39170 1104 6 1 52 1781666500 1781666529 REJECT OK
> 2026-06-17T03:21:40.000...	2 1579755500000 enl-Bafec76dffe930774 10.1.0.119 10.200.107.20 22 12835 6 2 176 1781666500 1781666529 ACCEPT OK

Activate Windows

Go to Settings to activate Windows.